

Root systems

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Chapter 1

Root pairings

1.1 Definitions

Recall that a *perfect pairing* is a quadruple $(R, M, N, \mathcal{L})$ where R is a ring, M and N are R -modules, and $\mathcal{L} : M \times N \rightarrow R$ is a bilinear map such that the induced maps $M \rightarrow \text{Hom}_R(N, R)$ and $N \rightarrow \text{Hom}_R(M, R)$ are isomorphisms of R -modules.

If I is a set, denote by $\text{Perm}(I)$ the group of permutations of I .

Definition 1. A *root pairing* is a tuple $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ where $(R, M, N, \mathcal{L})$ is a perfect pairing, I is a set, $\alpha : I \rightarrow M, i \mapsto \alpha_i$, $\alpha^\vee : I \rightarrow N, i \mapsto \alpha_i^\vee$, and $s : I \rightarrow \text{Perm}(I), i \mapsto s_i$, with the conditions that

1. $\alpha, \alpha^\vee$ are injective,
2. For all $i \in I$, $\mathcal{L}(\alpha_i, \alpha_i^\vee) = 2$,
3. For all $i, j \in I$, we have

$$\begin{aligned}\alpha_{s_i(j)} &= \alpha_j - \mathcal{L}(\alpha_j, \alpha_i^\vee) \alpha_i, \\ \alpha_{s_i(j)}^\vee &= \alpha_j^\vee - \mathcal{L}(\alpha_i, \alpha_j^\vee) \alpha_i^\vee.\end{aligned}$$

1.1.1 Properties

Definition 2. Fix a root pairing $\Phi = (R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$. We say that Φ is:

- *crystallographic* if for all $i, j \in I$,

$$\mathcal{L}(\alpha_i, \alpha_j^\vee) \in \mathbb{Z}.$$
- *reduced* if for all $i, j \in I$, α_i and α_j are linearly dependent over R if and only if $\alpha_i = \pm \alpha_j$.
- *finite* if I is finite.

Proposition 3. The root pairing defined in Theorem 14 is crystallographic, reduced and finite.

Proof. □

Definition 4. Given a root pairing $\Phi = (R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$, a set $\Delta \subset I$ is called a *base* of Φ if

1. The sets $\{\alpha_i : i \in \Delta\}$ and $\{\alpha_j^\vee : j \in \Delta\}$ are R -linearly independent, and
2. For every $i \in I$, there exist $k, \ell \in \mathbb{Z}_{>0}$, $a_1, \dots, a_k, b_1, \dots, b_\ell \in \Delta$ and $\varepsilon, \delta \in \{\pm 1\}$ such that

$$\alpha_j = \varepsilon \left(\sum_{\ell=1}^k \alpha_{a_\ell} \right) \quad \text{and} \quad \alpha_j^\vee = \delta \left(\sum_{\ell=1}^\ell \alpha_{b_\ell}^\vee \right).$$

Definition 5. A base is said to be *irreducible* if there exists $k \in \mathbb{Z}_{>0}$ and $a_1, \dots, a_k \in \Delta$ (not necessarily all distinct !) such that

1. $\{a_1, \dots, a_k\} = \Delta$, and
2. For every $i \in \{1, \dots, k-1\}$, $\mathcal{L}(\alpha_{a_i}, \alpha_{a_{i+1}}^\vee) \neq 0$.

Definition 6. Fix a root pairing $\Phi = (R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ and a base $\Delta \subset I$ of Φ . The pair (Φ, Δ) is said to be of

1. type *A* if there exists a sequence $(i_1, \dots, i_n)$, $n = |\Delta|$ of distinct elements of Δ such that

$$\mathcal{L}(\alpha_{i_k}, \alpha_{i_{k+1}}^\vee) = -1, \quad \text{for all } k \in \{1, \dots, n-1\}.$$

2. type *B* if there exists a sequence $(i_1, \dots, i_n)$, $n = |\Delta|$ of distinct elements of Δ such that

$$\mathcal{L}(\alpha_{i_k}, \alpha_{i_{k+1}}^\vee) = -1, \quad \text{for all } k \in \{1, \dots, n-2\}.$$

$$\mathcal{L}(\alpha_n, \alpha_{i_{n-1}}^\vee) = -2$$

3. type *C* if there exists a sequence $(i_1, \dots, i_n)$, $n = |\Delta|$ of distinct elements of Δ such that

$$\mathcal{L}(\alpha_{i_k}, \alpha_{i_{k+1}}^\vee) = -1, \quad \text{for all } k \in \{1, \dots, n-2\}$$

$$\mathcal{L}(\alpha_{n-1}, \alpha_{i_n}^\vee) = -2$$

4. type *D* if there exists distinct elements $d_1, d_2, t \in \Delta$ such that

$$\mathcal{L}(\alpha_{d_i}, \alpha_t^\vee) = -1$$

$$\Delta \setminus \{d_1, d_2\} \text{ is of type A}$$

5. type *E* if there exists distinct elements $d_1, d_2, t \in \Delta$ such that

$$\mathcal{L}(\alpha_{d_1}, \alpha_{d_2}^\vee) = -1, \mathcal{L}(\alpha_{d_2}, \alpha_t^\vee) = -1$$

$$\Delta \setminus \{d_1, d_2\} \text{ is of type A}$$

1.2 Root pairing of type A

1.2.1 Construction

Consider the perfect pairing $(R, M, N, \mathcal{L}) = (\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, (\cdot, \cdot))$ where $(\cdot, \cdot)$ is the canonical pairing between $\mathbb{Z}^n$ and its algebraic dual $(\mathbb{Z}^n)^* := \text{Hom}_{\mathbb{Z}}(\mathbb{Z}^n, \mathbb{Z})$, i.e. the dot product.

We denote by $e_1, \dots, e_n$ the standard basis of $\mathbb{Z}^n$ and by $e_1^*, \dots, e_n^*$ the dual basis of $(\mathbb{Z}^n)^*$.

Write $[1, n] = \{1, \dots, n\}$. An *interval* in $[1, n]$ is a subset of the form $[i, j] \stackrel{\text{denotes}}{=} \{i, i+1, \dots, j\}$ for some $1 \leq i \leq j \leq n$. We denote by I_n the set of intervals in $[1, n]$. Write $[i] := [i, i]$ for $1 \leq i \leq n$.

A *signed interval* in $[1, n]$ is an element of $I_n \times \{1, -1\}$. We denote by I_n^+ the set of signed intervals of the form $(J, 1)$ and by I_n^- the set of signed intervals of the form $(J, -1)$. By abuse of notation, we identify I_n with I_n^+ and write J for the signed interval $(J, 1)$ for all $J \in I_n$. If $J \in I_n$ then we write $-J$ for the signed interval $(J, -1)$.

Let $\mathcal{A}_n = (a_{i,j})$ be the $n \times n$ integer matrix defined by

- i) $a_{i,i} = 2$ for all $i \in [1, n]$,
- ii) $a_{i,j} = -1$ if $|i - j| = 1$, and
- iii) $a_{i,j} = 0$ otherwise.

The matrix $\mathcal{A}_n$ is often called a *Cartan matrix of type A_n* . In the rest of this subsection we write $A = \mathcal{A}_n$.

Lemma 7. *For any $n \in \mathbb{Z}_{\geq 1}$, $\det(\mathcal{A}_n) = n + 1$. In particular, $\mathcal{A}_n$ is invertible.*

Proof. By a direct computation, $\det(\mathcal{A}_1) = 2$ and $\det(\mathcal{A}_2) = 3$. Now fix $n \geq 3$, and suppose that the claim holds for all $k < n$. Using the Laplace expansion along the first row, we have

$$\begin{aligned} \det(\mathcal{A}_n) &= 2 \det(\mathcal{A}_{n-1}) - \det(\mathcal{A}_{n-2}) \\ &= 2n - (n-1) = n + 1. \end{aligned}$$

The statement follows. □

Corollary 8. *The set $\{\mathcal{A}_n e_k : 1 \leq k \leq n\}$ is linearly independent over $\mathbb{Q}$.*

Proof. If $\sum_{k=1}^n c_k \mathcal{A}_n e_k = 0$ for some $c_1, \dots, c_n \in \mathbb{Q}$, then $\mathcal{A}_n (\sum_{k=1}^n c_k e_k) = 0$. Left-multiplying both sides by $\mathcal{A}_n^{-1}$, we have $\sum_{k=1}^n c_k e_k = 0$. Since $\{e_k : 1 \leq k \leq n\}$ is a basis of $\mathbb{Z}^n$, we have $c_k = 0$ for all k . □

Define

$$\alpha : I_n \times \{\pm 1\} \rightarrow \mathbb{Z}^n, \quad \forall J \in I_n, \quad \alpha_{\pm J} = \pm \left(\sum_{j \in J} A e_j \right), \quad (1.1)$$

$$\alpha^\vee : I_n \times \{\pm 1\} \rightarrow (\mathbb{Z}^n)^*, \quad \forall J \in I_n, \quad \alpha_{\pm J}^\vee = \pm \left(\sum_{j \in J} e_j^* \right). \quad (1.2)$$

Lemma 9. *The maps $\alpha : I_n \times \{\pm 1\} \rightarrow \mathbb{Z}^n$ and $\alpha^\vee : I_n \times \{\pm 1\} \rightarrow (\mathbb{Z}^n)^*$ are injective.*

Proof. Let $J, K \in I_n \times \{\pm 1\}$. Write $J = \varepsilon J_0$ and $K = \delta K_0$, where $J_0, K_0 \in I_n$ and $\varepsilon, \delta \in \{\pm 1\}$. Suppose first that $\alpha_J = \alpha_K$. Then

$$\sum_{r=1}^n (\varepsilon \mathbf{1}_{r \in J_0} - \delta \mathbf{1}_{r \in K_0}) A e_r = 0.$$

Since the vectors $Ae_1, \dots, Ae_n$ are linearly independent, each coefficient is zero. Thus

$$\varepsilon \mathbf{1}_{r \in J_0} = \delta \mathbf{1}_{r \in K_0} \quad \text{for all } r \in [1, n].$$

It follows that $r \in J_0$ if and only if $r \in K_0$, so $J_0 = K_0$; since these intervals are nonempty, the same equality of coefficients also gives $\varepsilon = \delta$. Hence $J = K$.

The proof for $\alpha^\vee$ is identical, using the linear independence of the dual basis $e_1^*, \dots, e_n^*$. Indeed, if $\alpha_J^\vee = \alpha_K^\vee$, then

$$\sum_{r=1}^n (\varepsilon \mathbf{1}_{r \in J_0} - \delta \mathbf{1}_{r \in K_0}) e_r^* = 0,$$

and the same coefficient comparison gives $J_0 = K_0$ and $\varepsilon = \delta$. □

Lemma 10. *For every $J \in I_n \times \{\pm 1\}$, $\mathcal{L}(\alpha_J, \alpha_J^\vee) = 2$.*

Proof. Write $J = \varepsilon J_0$, where $J_0 = [p, q]$ and $\varepsilon \in \{\pm 1\}$. Since $\varepsilon^2 = 1$,

$$\begin{aligned} \mathcal{L}(\alpha_J, \alpha_J^\vee) &= \sum_{u=p}^q \sum_{v=p}^q e_v^*(Ae_u) \\ &= \sum_{u=p}^q \sum_{v=p}^q a_{v,u}. \end{aligned}$$

If $m = q - p + 1$ is the length of J_0 , this sum has m diagonal terms, each equal to 2, and $2(m-1)$ ordered adjacent off-diagonal terms, each equal to -1 . All other terms are zero. Hence

$$\mathcal{L}(\alpha_J, \alpha_J^\vee) = 2m - 2(m-1) = 2.$$

□

We first define $s_J(K)$ for $J, K \in I_n$. Set $L = (J \cup K) \setminus (J \cap K)$. It is clear that J, K, L satisfy exactly one of the following four conditions.

R1. $L = \emptyset$.

R2. $L \in I_n$ and $K \subset J$,

R3. $L \in I_n$ and $K \not\subset J$,

R4. $L \notin I_n \cup \{\emptyset\}$.

Then,

$$s_J(K) = \begin{cases} -K & \text{in case \textbf{R1},} \\ -L & \text{in case \textbf{R2},} \\ L & \text{in case \textbf{R3},} \\ K & \text{in case \textbf{R4}.} \end{cases} \quad (1.3)$$

If $J = [i, j]$ and $K = [k, l]$, then (1.3) can be rewritten as

$$s_{[i,j]}[k, l] = \begin{cases} -[k, l] & \text{if } (i, j) = (k, l) \\ -[l+1, j] & \text{if } i = k, j > l \\ [j+1, l] & \text{if } i = k, j < l \\ -[i, k-1] & \text{if } i < k, j = l \\ [k, i-1] & \text{if } i > k, j = l \\ [k, j] & \text{if } i = l+1 \\ [i, l] & \text{if } k = j+1 \\ [k, l] & \text{otherwise.} \end{cases} \quad (1.4)$$

We extend s_J to $I_n \times \{\pm 1\}$ by setting $s_J(-K) = -s_J(K)$ for all $K \in I_n$. We extend s to $I_n \times \{\pm 1\}$ by setting $s_{-J} = s_J$ for all $J \in I_n$.

Lemma 11. *For all $J \in I_n \times \{\pm 1\}$, we have $s_J \in \text{Perm}(I_n \times \{\pm 1\})$.*

Proof. Fix $J \in I_n \times \{\pm 1\}$. We prove that $s_J^2 = \text{id}$ by a straightforward case-by-case check using the definition of s_J and the four cases above. Then $s_J^{-1} = s_J$, and our claim is proved. $\square$

Lemma 12. *For all $J, K \in I_n \times \{\pm 1\}$,*

$$\alpha_{s_J(K)} = \alpha_K - \mathcal{L}(\alpha_K, \alpha_J^\vee) \alpha_J.$$

Proof. First assume that $J = [i, j]$ and $K = [k, l]$ are positive intervals. Let $c = \mathcal{L}(\alpha_K, \alpha_J^\vee)$. For $u \in [1, n]$,

$$\sum_{r=k}^l a_{u,r} = \mathbf{1}_{u=k} + \mathbf{1}_{u=l} - \mathbf{1}_{u=k-1} - \mathbf{1}_{u=l+1}.$$

Summing this identity over $u = i, \dots, j$ gives

$$c = \delta_{i,k} + \delta_{j,l} - \delta_{i,l+1} - \delta_{j,k-1},$$

where δ denotes the Kronecker delta. Comparing this value with (1.4), we get the following cases:

condition	c	$\alpha_K - c\alpha_J$
$(i, j) = (k, l)$	2	$-\alpha_{[k,l]}$
$i = k, j > l$	1	$-\alpha_{[l+1,j]}$
$i = k, j < l$	1	$\alpha_{[j+1,l]}$
$i < k, j = l$	1	$-\alpha_{[i,k-1]}$
$i > k, j = l$	1	$\alpha_{[k,i-1]}$
$i = l+1$	-1	$\alpha_{[k,j]}$
$k = j+1$	-1	$\alpha_{[i,l]}$
otherwise	0	$\alpha_{[k,l]}$

Each entry in the last column is exactly $\alpha_{s_J(K)}$ by (1.4). Thus the desired identity holds for positive intervals.

Now write arbitrary signed intervals as $J = \varepsilon J_0$ and $K = \delta K_0$, with $J_0, K_0 \in I_n$ and $\varepsilon, \delta \in \{\pm 1\}$. Since $s_{\varepsilon J_0}(\delta K_0) = \delta s_{J_0}(K_0)$ and $\mathcal{L}(\alpha_K, \alpha_J^\vee) = \delta \varepsilon \mathcal{L}(\alpha_{K_0}, \alpha_{J_0}^\vee)$, the positive case

gives

$$\begin{aligned}\alpha_{s_J(K)} &= \delta \alpha_{s_{J_0}(K_0)} \\ &= \delta(\alpha_{K_0} - \mathcal{L}(\alpha_{K_0}, \alpha_{J_0}^\vee) \alpha_{J_0}) \\ &= \alpha_K - \mathcal{L}(\alpha_K, \alpha_J^\vee) \alpha_J.\end{aligned}$$

□

Lemma 13. *For all $J, K \in I_n \times \{\pm 1\}$,*

$$\alpha_{s_J(K)}^\vee = \alpha_K^\vee - \mathcal{L}(\alpha_J, \alpha_K^\vee) \alpha_J^\vee.$$

Proof. First assume that $J = [i, j]$ and $K = [k, l]$ are positive intervals, and set $c = \mathcal{L}(\alpha_J, \alpha_K^\vee)$. Since A is symmetric, the endpoint computation in the proof of Lemma 12 gives the same value

$$c = \delta_{i,k} + \delta_{j,l} - \delta_{i,l+1} - \delta_{j,k-1}.$$

The case table in that proof used only additivity of sums over intervals. Replacing each Ae_r by e_r^* in the last column gives

$$\alpha_{s_J(K)}^\vee = \alpha_K^\vee - c \alpha_J^\vee$$

for positive intervals.

For general signed intervals, write $J = \varepsilon J_0$ and $K = \delta K_0$, with $J_0, K_0 \in I_n$ and $\varepsilon, \delta \in \{\pm 1\}$. Using $s_{\varepsilon J_0}(\delta K_0) = \delta s_{J_0}(K_0)$ and $\mathcal{L}(\alpha_J, \alpha_K^\vee) = \varepsilon \delta \mathcal{L}(\alpha_{J_0}, \alpha_{K_0}^\vee)$, the positive case gives

$$\begin{aligned}\alpha_{s_J(K)}^\vee &= \delta \alpha_{s_{J_0}(K_0)}^\vee \\ &= \delta(\alpha_{K_0}^\vee - \mathcal{L}(\alpha_{J_0}, \alpha_{K_0}^\vee) \alpha_{J_0}^\vee) \\ &= \alpha_K^\vee - \mathcal{L}(\alpha_J, \alpha_K^\vee) \alpha_J^\vee.\end{aligned}$$

□

Theorem 14. *Fix $n \in \mathbb{Z}_{>0}$, $R = \mathbb{Z}$, $M = \mathbb{Z}^n$ and $N = (\mathbb{Z}^n)^*$. Let $\mathcal{L}$ be the canonical pairing, let $I = I_n \times \{\pm 1\}$ be the set of signed intervals, let α and $\alpha^\vee$ be as in (1.1) and (1.2) respectively, and let s be as in (1.3). Then the 8-tuple $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$ is a root pairing.*

Proof. The quadruple $(\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, \mathcal{L})$ is the standard perfect pairing between a finite free $\mathbb{Z}$ -module and its dual. It remains to verify the three extra conditions in Definition 1.

The maps α and $\alpha^\vee$ are injective by Lemma 9. For every signed interval $J \in I$, Lemma 10 gives $\mathcal{L}(\alpha_J, \alpha_J^\vee) = 2$. For every $J \in I$, Lemma 11 gives $s_J \in \text{Perm}(I)$. Finally, for all $J, K \in I$, Lemmas 12 and 13 give

$$\alpha_{s_J(K)} = \alpha_K - \mathcal{L}(\alpha_K, \alpha_J^\vee) \alpha_J \quad \text{and} \quad \alpha_{s_J(K)}^\vee = \alpha_K^\vee - \mathcal{L}(\alpha_J, \alpha_K^\vee) \alpha_J^\vee.$$

These are precisely the axioms in Definition 1, so the stated 8-tuple is a root pairing. □

Proposition 15. *Let Φ denote the root pairing constructed in Theorem 14, and set $\Delta = \{[1], [2], \dots, [n]\}$. Then Δ is an irreducible base of Φ , and the pair (Φ, Δ) is of type A.*

Proof. □